06 Pathway enrichment (Chapter 3)

Each fitted contrast is tested by preranked gene-set enrichment (fgsea), ranking genes by the limma moderated t-statistic.

The gene-set lib is built from:

- two curated PI3K–mTOR sets defined below;
- Hallmark,
- KEGG mTOR/insulin/neurotrophin,
- Reactome mTOR and
- HPO epilepsy/cortical sets in MSigDB v2024.1;
- a separate HPO epilepsy and cortical-malformation GMT.

For `HP_CORTICAL_DYSPLASIA` and `HP_FOCAL_CORTICAL_DYSPLASIA`, the version in the separate GMT replaces the MSigDB version. Download links are included in README.

```
suppressPackageStartupMessages(library(fgsea))
source(file.path("..", "scripts", "helpers.R"))
paths <- load_paths()
```

Parameters

```
collection_patterns <- list(
  hallmark = "^HALLMARK_",
  kegg = "^KEGG_MTOR_SIGNALING|^KEGG_INSULIN_SIGNALING|^KEGG_NEUROTROPHIN",
  reactome = "MTOR",
  hpo = "^HP_.*EPILEPSY|^HP_.*CORTICAL")
prefer_newer_hpo <- c("HP_CORTICAL_DYSPLASIA", "HP_FOCAL_CORTICAL_DYSPLASIA")

fgsea_seed <- 12345L
min_size <- 15L
max_size <- 500L
min_ranked_genes <- 100L
fdr_threshold <- 0.05
```

Gene sets

```
gene_sets <- list(
  PIK3CA_PROXIMAL_CASCADE = c(
    "PIK3CA", "PIK3CB", "PIK3CD", "PIK3CG", "PIK3R1", "PIK3R2", "PIK3R3", "PTEN",
    ↪  "PDPK1",
    "AKT1", "AKT2", "AKT3", "TSC1", "TSC2", "RHEB", "MTOR", "RPTOR", "MLST8", "AKT1S1",
    "RPS6KB1", "RPS6KB2", "EIF4EBP1"),

  NEURON_MTOR_RAPAMYCIN_RESP = c(
    "EEF1A1", "EEF1A2", "EEF2", "EIF4A1", "EIF4B", "EIF4E", "RPS6", "RPS6KB1", "RPS6KB2",
    "RPTOR", "NPM1", "NOP56", "NOP58", "FBL", "DKC1", "ARC", "CAMK2A", "GRIA1", "BDNF",
    "MAP1B", "ULK1", "ATG13", "RB1CC1", "BECN1", "MYC", "CCND1", "CDK4", "LDHA", "PKM",
```

```
    "HK2", "SLC2A1", "DDIT4", "SESN1", "SESN2"))

gmt_files <- list.files(paths$msigdb_dir, pattern = "\\.symbols\\.gmt$", full.names =
 ↪  TRUE)

for (f in gmt_files) {
  g <- fgsea::gmtPathways(f)
  keep <- unique(unlist(lapply(collection_patterns, grep, x = names(g), value = TRUE,
 ↪  ignore.case = TRUE), use.names = FALSE))
  gene_sets <- c(gene_sets, g[setdiff(keep, prefer_newer_hpo)])
}

epi <- fgsea::gmtPathways(paths$epilepsy_gmt)
epi <- lapply(epi, function(x) x[!grepl("^http", x)])
gene_sets <- c(gene_sets, epi[lengths(epi) > 0L])

# Retain the first copy of any duplicate gene set names

dup_names <- unique(names(gene_sets)[duplicated(names(gene_sets))])
keep <- rep(TRUE, length(gene_sets))
for (nm in dup_names) {
  idx <- which(names(gene_sets) == nm)
  for (i in seq_along(idx)) {
    if (!keep[idx[i]] || i == length(idx)) next
    for (j in idx[(i + 1L):length(idx)])
      if (keep[j] && setequal(unique(gene_sets[[idx[i]]]), unique(gene_sets[[j]])))
       ↪  keep[j] <- FALSE
  }
}

gene_sets <- gene_sets[keep]
length(gene_sets)
```

```
## [1] 112
```

fgsea per contrast

```
de_dir  <- file.path(paths$work_dir, "05_differential_expression")

out_dir <- file.path(paths$work_dir, "06_pathways")

for (f in list.files(de_dir, pattern = "_toptable\\.tsv$", recursive = TRUE, full.names =
 ↪  TRUE)) {
  cell_type   <- basename(dirname(f))
  contrast_id <- sub("_toptable\\.tsv$", "", basename(f))
  de <- utils::read.delim(f, stringsAsFactors = FALSE, check.names = FALSE)

  stat <- setNames(de$t, de$gene_id)
  ranks <- sort(stat[is.finite(stat)], decreasing = TRUE)
  if (length(ranks) < min_ranked_genes) next

  set.seed(fgsea_seed)

  fg <- as.data.frame(fgsea::fgsea(pathways = gene_sets, stats = ranks, minSize =
 ↪  min_size,
```

```
                                    maxSize = max_size, eps = 0, nproc = 1L))
  fg$cell_type <- cell_type; fg$contrast_id <- contrast_id
  fg$padj_significant <- is.finite(fg$padj) & fg$padj < fdr_threshold

  leading_edge <- data.frame(cell_type = cell_type, contrast_id = contrast_id,
                             pathway_id = fg$pathway, NES = fg$NES, padj = fg$padj,
                             leading_edge_genes = vapply(fg$leadingEdge, paste, "",
                              ↪  collapse = ";"),
                             gene_membership_count = lengths(fg$leadingEdge))

  fg$leadingEdge <- vapply(fg$leadingEdge, paste, "", collapse = ";")

  tag <- paste0(cell_type, "__", contrast_id)
  save_tsv(fg, file.path(out_dir, paste0("fgsea_", tag, ".tsv")))
  save_tsv(leading_edge, file.path(out_dir, paste0("leading_edge_", tag, ".tsv")))
}
```

```
## Warning in prepareStats(stats, scoreType, gseaParam): There are ties in the preranked
 ↪  stats (0.03% of the list).
## The order of those tied genes will be arbitrary, which may produce unexpected results.
```

Check against uploaded results

```
dep_dir <- file.path(paths$zenodo_root, "results", "pathways")

for (f in list.files(out_dir, "^fgsea_.*\\.tsv$")) {
  a <- read.delim(file.path(out_dir, f)); b <- read.delim(file.path(dep_dir, f))
  stopifnot(identical(a$pathway, b$pathway),
            isTRUE(all.equal(a$NES, b$NES, tolerance = 1e-8)),
            isTRUE(all.equal(a$padj, b$padj, tolerance = 1e-8)),
            identical(a$leadingEdge, b$leadingEdge))
}

sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.
 ↪  framework/Versions/A/libBLAS.dylib
## LAPACK:
 ↪  /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
 ↪  LAPACK version 3.12.1
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats4    stats     graphics  grDevices utils     datasets  methods
```

```
## [8] base
##
## other attached packages:
##  [1] fgsea_1.39.4                edgeR_4.8.2
##  [3] limma_3.66.0                igraph_2.2.1
##  [5] scater_1.38.0               ggplot2_4.0.1
##  [7] scran_1.38.0                Matrix_1.7-4
##  [9] scuttle_1.20.0              SingleCellExperiment_1.32.0
## [11] SummarizedExperiment_1.40.0 Biobase_2.70.0
## [13] GenomicRanges_1.62.1        Seqinfo_1.0.0
## [15] IRanges_2.44.0              S4Vectors_0.48.0
## [17] BiocGenerics_0.56.0         generics_0.1.4
## [19] MatrixGenerics_1.22.0       matrixStats_1.5.0
##
## loaded via a namespace (and not attached):
##  [1] fastmatch_1.1-8     beeswarm_0.4.0      gtable_0.3.6
##  [4] xfun_0.55           ggrepel_0.9.6       lattice_0.22-7
##  [7] vctrs_0.7.1         tools_4.5.2         parallel_4.5.2
## [10] tibble_3.3.1        cluster_2.1.8.2     pkgconfig_2.0.3
## [13] BiocNeighbors_2.4.0 data.table_1.18.0   RColorBrewer_1.1-3
## [16] S7_0.2.1            dqrng_0.4.1         lifecycle_1.0.5
## [19] compiler_4.5.2      farver_2.1.2        statmod_1.5.1
## [22] tinytex_0.58        bluster_1.20.0      codetools_0.2-20
## [25] vipor_0.4.7         htmltools_0.5.9     yaml_2.3.12
## [28] pillar_1.11.1       BiocParallel_1.44.0 DelayedArray_0.36.0
## [31] viridis_0.6.5       abind_1.4-8         tidyselect_1.2.1
## [34] rsvd_1.0.5          locfit_1.5-9.12     metapod_1.18.0
## [37] digest_0.6.39       BiocSingular_1.26.0 dplyr_1.2.0
## [40] cowplot_1.2.0       fastmap_1.2.0       grid_4.5.2
## [43] cli_3.6.5           SparseArray_1.10.8  magrittr_2.0.4
## [46] S4Arrays_1.10.1     withr_3.0.2         scales_1.4.0
## [49] ggbeeswarm_0.7.3    rmarkdown_2.30      XVector_0.50.0
## [52] otel_0.2.0          gridExtra_2.3       ScaledMatrix_1.18.0
## [55] beachmat_2.26.0     evaluate_1.0.5      knitr_1.51
## [58] viridisLite_0.4.2   irlba_2.3.7         rlang_1.1.7
## [61] Rcpp_1.1.1          glue_1.8.0          R6_2.6.1
```